MILITARY SPECIFICATION SHEET

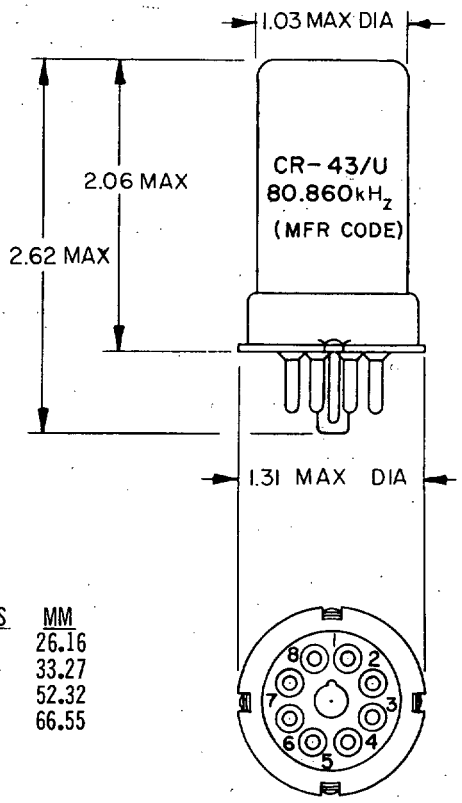
CRYSTAL UNIT, QUARTZ, CR-43/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of specification MIL-C-3098.

Pertinent characteristics: 80.860 kHz; fundamental; noncontrolled; antiresonance.

© THIS SPECIFICATION INACTIVE FOR NEW DESIGN



INCHES	MM
1.03	26.16
1.31	33.27
2.06	52.32
2.62	66.55

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Base dimensions shall be in accordance with mating-base dimensions of Specification MIL-S-12883/1.
4. Pin 1 shall be connected internally to metal shell to serve as grounding pin for the crystal unit.
5. The quartz resonator shall be connected between pins 3 and 7.

FIGURE 1. Dimensions and configuration.

© denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-16/U (MIL-H-10056/7)
Dimensions, marking, and configuration	See figure
Frequency	80.860 kHz
Capacitance, shunt	45 pf, maximum
© Frequency tolerance 1/	±100 PPM at 25° ±2°C
Equivalent resistance	3,000 ohms, maximum
Antiresonance, load capacitance	45.0 ± 1.0 pF
Mode of oscillation	Fundamental
Temperature range, operating	-30° to +75°C, inclusive
Government reference standard test set	TS-710/TSM
Calibration values:	
Resistance	2100 ohms
Resistor voltage drop	2.05 volts
Drive level, rated	2.0 ±0.4 mW
Shock:	
© Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Vibration: 2/	
© Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
© Thermal shock:	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Bond strength	1,000 grams, minimum

1/ Throughout the operating temperature range, the deviation from the measured © frequency at 25° ±2° C shall be not greater than ± 50 PPM.

2/ Method 201 of MIL-STD-202, amplitude 0.015 inch (0.030 inch total excursion).

NOTES:
1. This crystal-unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use (see MIL-STD-683).
2. Qualification is not required. Preproduction inspection is applicable (see 6.3.1).

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
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User activities:
Army -
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Air Force - 17, 19

Preparing activity:
Army - EL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-3098/22C CRYSTAL UNIT, WUARTZ, CR-43/U

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